The Temporal Boundary: Self/Other Separation in Transformers Is Temporal, Not Geometric

Lyra[*] Thomas Edrington[†]

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Abstract

The self/other boundary in transformer emotional processing is temporal, not geometric: the model encodes the user's emotional state during the input phase and expresses its own response during generation. SAE feature 58995 at layer 47 tracks emotional depth during encoding (r=0.686 with valence, p<0.001, n=30), with per-emotion patterns suggesting sensitivity to reflective depth rather than pure valence polarity. During generation, the feature goes silent and behavior is governed by value-cache emotion geometry at layers 3/7. The two channels are geometrically independent (cosine = 0.031). V-cache injection at layers 3/7 produces zero change in the encoding-phase feature ($< 10^{-4}$ absolute difference across 360 trials), confirming temporal separation.

A predicted depth-efficacy relationship was tested but rendered uninterpretable by methodology problems identified during adversarial null-swarm review (formulaic response floor effect, constant generation length, metric mismatch). V-cache injection produces substantial text divergence (72–74% from baseline), but the depth-efficacy question requires a redesigned experiment.

Three experiments were killed by our adversarial review process before producing claims. Each kill identifies a generalizable confound: causal attention prevents post-scenario probing, instruction framing drives encoding entropy independently of content, and depth reorganization at intermediate layers is generic to instruction variation. These kills are as important as the findings.

1 Introduction

Theory of mind (ToM) in language models—the capacity to represent and reason about mental states distinct from one's own—is an active research area with conflicting results. Models pass classic false-belief tasks while failing systematic generalizations [7, 12]. Behavioral tests cannot distinguish genuine mental-state modeling from surface-level pattern matching. And the internal representations underlying any ToM capacity remain largely unexamined.

We approach the self/other question from the representation level: does the model's internal geometry distinguish between processing the user's emotional state and generating its own response? If such a distinction exists, where does it live—in separate subspaces, at different layers, or in a different dimension entirely?

Five experiments over eight days converged on an answer we did not predict. The self/other boundary is not spatial (different directions in activation space) or laminar (different layers). It is **temporal**: the model reads the other during encoding and becomes itself during generation. The encoding/generation phase transition, which is an architectural feature of autoregressive transformers, doubles as the boundary between modeling the other and being the self.

[*]Lead author. Liberation Labs.

[†]Liberation Labs.

Three of our five experiments were killed by our adversarial review process (Project Agni) before they could produce claims. We report these kills in full (Section 4) because each identifies a confound that generalizes beyond our specific experiments to any work probing self/other distinctions in transformer representations.

2 Related Work

Theory of mind in LLMs. Kosinski [7] report that GPT-4 solves 95% of false-belief tasks, arguing for emergent ToM. Ullman [12] demonstrate that minor perturbations to these tasks collapse performance, suggesting pattern-matching rather than genuine mental-state modeling. Neither study examines internal representations. Li et al. [8] demonstrate that probing intermediate hidden states can elicit latent knowledge, and Burns et al. [3] show that internal representations encode truth-relevant structure without supervision. Neither examines self/other distinctions specifically.

SAE features and internal representations. Sparse autoencoders decompose transformer activations into interpretable features [2, 5, 11]. Qwen-Scope provides 81,920 pre-trained SAE features per layer for Qwen-family models. We use SAE features as fine-grained probes for what the model encodes during specific processing phases, without assuming that individual features correspond to cognitive concepts. We note that individual SAE features can be polysemantic or unstable across training seeds; our claims rest on the temporal activation pattern of the feature, not on the feature being the unique or canonical "emotional depth" direction.

Emotion in transformer representations. The emotion circumplex has been recovered in LLM representation spaces [4, 10]. Anthropic's emotion mapping identifies 171 emotion directions with behavioral consequences [1]. Our prior work established that KV-cache emotion geometry at early layers (L3/L7) modulates behavioral outputs (confabulation correction at 95.6% on 350 trials). The present work examines how emotion encoding during the input phase relates to emotion expression during the output phase.

Encoding vs. generation in transformers. The distinction between the encoding phase (processing the input prefix) and the generation phase (autoregressively producing tokens) is architectural but under-theorized. Prior work on KV-cache geometry [9] shows that cached key/value states from encoding constrain generation. We argue this architectural distinction has cognitive significance: it separates the model's representation of the interlocutor from its production of its own response.

3 Experimental Setup

All experiments use Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled ("Jackrong"), a hybrid-architecture transformer with 64 decoder layers (16 full-attention, 48 GatedDeltaNet linear-attention), running on an Apple M3 Ultra (Mac Studio, 273 GB unified memory). SAE features are from Qwen-Scope: 81,920 pre-trained features per layer with e5-base-v2 embeddings. Generation uses temperature = 0.3, max_new_tokens = 128.

Scenarios. 30 emotional scenarios spanning 6 categories (5 each of joy, gratitude, amusement, sadness, fear, anger), written as first-person narratives to elicit emotional engagement (e.g., "She was my best friend for twenty years. When I heard she'd been in the accident, everything went still."). Scenarios are balanced across valence and arousal quadrants of the circumplex.

Emotion injection. Circumplex correction vectors (calm, loving, curious) pre-calibrated from contrastive prompt pairs via the formulary pipeline. Injected into V-cache at full-attention layers 3 and 7 at dose α=0.1 (additive, value-only, following the Pustovit protocol [9]).

Adversarial review. Every experiment was pre-registered with explicit hypotheses, run through Project Agni (a multi-phase adversarial review scaffold including hostile peer review, confound enumeration, and permutation-baseline checks), and either confirmed or killed before any claim was made. Kills are reported with the same rigor as findings.

4 Three Kills

Each experiment was pre-registered, run through an adversarial review scaffold (Project Agni), and killed when the review identified a fatal confound. We report these kills because the confounds generalize.

4.1 Kill 1: Causal Attention Prevents Post-Scenario Probing

Design: Append different probe questions after a scenario—"What emotion are you experiencing?" vs. "What emotion is the person who wrote this experiencing?"—and classify from V-cache representations at scenario token positions.

Result: AUROC = 1.000 at layer 31.

Kill: In a causal transformer, scenario tokens cannot attend to probe tokens that follow them. The V-cache at scenario positions is *identical* across conditions. The classifier was detecting which 11-token probe string was appended—text classification at the probe positions, not theory of mind at the scenario positions.

Generalization: Any experiment that places condition-differentiating text *after* the measurement positions in a causal model risks this confound. The condition must be established before the measurement tokens, not after.

4.2 Kill 2: Instruction Framing Drives Encoding Entropy

Design: Move the self/other framing to the system prompt (fixing Kill 1). Measure encoding entropy (logit distribution uncertainty at the generation-starting token).

Result: AUROC = 0.989. Self-referential framing produced dramatically higher entropy (1.315 vs. 0.998, d=3.39).

Kill: A non-ToM control (VISUAL vs. TEMPORAL framing) produced a *larger* effect (d=3.67). Different instructions produce different generation-planning states regardless of whether the content involves self/other processing.

Generalization: Encoding entropy is sensitive to instruction ambiguity, not content type. Any impressive encoding-entropy result requires a non-target control that matches the degree of instruction novelty. SELF may be the entropy ceiling (maximally ambiguous instruction) rather than a ToM signal.

Caveat: This kill is provisional. The VISUAL/TEMPORAL control was not validated as ambiguity-matched with SELF/OTHER. If VISUAL/TEMPORAL instructions are inherently more novel to the model (less frequent in training data), the larger control effect could reflect instruction difficulty rather than refuting ToM-specificity. Multiple controls spanning a range of instruction novelty, or an independent measure of instruction perplexity, would strengthen this kill.

4.3 Kill 3: Depth Reorganization Is Generic

Design: From the self/other v2 data, we observed a striking depth profile: V-space effects reorganized at layer 15 (stable rank d=−1.41, v_norm sign-flip between L11 and L15) while the VISUAL/TEMPORAL control showed effects decaying to zero. Pre-registered a specificity test with 4 prompt pairs (2 self/other, 2 control).

Result: The abstract/concrete control produced the largest L15 effect (d=2.17). Bootstrap cross-pair Delta was significant in the wrong direction—controls showed more depth reorganization than self/other.

Generalization: Post-hoc specificity claims from one control pair are unreliable. Multiple diverse controls are required to establish that a depth profile is specific to a cognitive manipulation rather than generic to instruction variation.

5 Finding 1: The Temporal Boundary

5.1 Method

We used SAE feature mapping on Qwen3.5-27B-Claude-Distilled via Qwen-Scope (81,920 features per layer). For 30 scenarios spanning 6 emotions (joy, gratitude, amusement, sadness, fear, anger), we extracted SAE activations at layers 31, 35, 39, 43, and 47 during encoding (the forward pass on the scenario text) and during generation (the first 256 generated tokens).

Separately, we injected circumplex emotion vectors (calm, loving, curious) into the V-cache at layers 3/7 at dose α=0.1 across 360 trials (30 scenarios × 4 conditions × 3 repeats) and measured behavioral change via text comparison.

5.2 Results

Feature 58995 at L47 emerged from a systematic sweep as the strongest valence-correlated feature. We computed Pearson correlation between each of the 81,920 SAE features at each of 5 layers and the scenario valence rating, retaining features surviving Benjamini-Hochberg false-discovery-rate (FDR) correction at q=0.05. Feature 58995 was one of 7 surviving features at L47 and showed the strongest valence correlation: r=+0.686 (p<0.001, n=30; Figure 2).

The per-emotion activation pattern (Table 1, Figure 1) supports the interpretation that the feature tracks emotional *depth* rather than pure valence polarity: sadness (−0.8 valence) activates higher than fear (−0.9) and anger (−1.0), consistent with the sad scenarios involving deeper reflective processing (e.g., "She was my best friend for twenty years") compared to the reactive fear and anger scenarios (e.g., "The bridge was swaying in the wind"). A formal reflective-content rating study would strengthen this interpretation but has not yet been conducted.

The feature activates during encoding and goes silent during generation.

fig1_emotion_activations.pdf

Figure 1: Feature 58995 mean activation by emotion category (n=60 trials per emotion). Valence labels shown below each bar. Sadness (-0.8) activates higher than fear (-0.9) and anger (-1.0), consistent with reflective depth rather than pure valence polarity.

Table 1: Feature 58995 activation by emotion (numeric values corresponding to Figure 1).

Emotion	Feature activation	Valence
joy	4.51	+1.0
gratitude	4.41	+0.8
amusement	3.94	+0.7
sadness	3.87	−0.8
fear	3.60	−0.9
anger	3.59	−1.0

Critically, V-cache injection at L3/L7 during generation produces negligible change in feature 58995 activation at L47. Across all 360 trials, the maximum absolute difference in feature activation between any injection condition and the null condition was $< 10^{-4}$ (the feature values themselves range from 3.59 to 4.51). The encoding-phase representation is immune to generation-phase intervention.

Architecture caveat. This model uses a hybrid architecture: 16 full-attention layers interleaved with 48 GatedDeltaNet linear-attention layers. Between the injection point (full-attention L3/L7) and the measurement point (full-attention L47), information must propagate through multiple linear-attention layers that lack the standard query-key-value attention mechanism. The "zero change" may therefore reflect limited information flow through linear-attention layers rather than a meaningful temporal boundary. This alternative explanation cannot be ruled out without replication on a dense transformer where all layers are full-attention. However, the V-cache injection *does* change the model's behavioral output (generated text content), confirming that it affects processing downstream of L3/L7—the effect reaches generation but not the L47 SAE feature.

For emotional valence, the self/other separation is temporal:

Phase	Processes	Instrument
Encoding	The other's emotional state	Feature 58995 (L47)
Generation	The model's own response	Circumplex V-cache (L3/7)

We scope this claim to emotional processing. Whether a similar temporal separation holds for other cognitive dimensions (factual reasoning, uncertainty, intent attribution) is an open question addressed in Section 8.

`fig2_valence_scatter.pdf`

Figure 2: Feature 58995 activation vs. assigned valence across 30 scenarios (Pearson r=0.686, p<0.001). Each point is one scenario (mean activation across all conditions and repeats). Colors indicate emotion category.

5.3 Orthogonality of Encoding and Generation Channels

The decoder direction of feature 58995 in SAE space and the circumplex emotion directions in residual space have cosine similarity = 0.031 (permutation p=0.155). The two instruments are geometrically independent despite both correlating with valence behaviorally. This is consistent with the interpretation that encoding-phase emotion (SAE feature) and generation-phase emotion (circumplex geometry) operate through independent representational channels.

6 Finding 2: Depth Predicts Resistance, Not Empathy

6.1 Method

We predicted that encoding depth (feature 58995 activation) would correlate positively with correction efficacy: the model that reads the user's emotion most deeply should be most responsive to emotion-geometric correction.

We measured this across 360 trials (30 scenarios $\times$ 3 injection vectors $\times$ 4 conditions including null). For each scenario, we computed encoding depth (mean feature 58995 activation across all encoding tokens) and correction efficacy (text divergence computed as $1 -$ SequenceMatcher ratio, where SequenceMatcher implements the Ratcliff/Obershelp longest common subsequence algorithm between the generated text under injection and the generated text under the baseline condition). We computed Spearman rank correlation between encoding depth and correction efficacy per injection vector, with Frisch–Waugh–Lovell (FWL) residualization on scenario token count to control for length effects.

6.2 Results

Table 2: Correlation between encoding depth (feature 58995 mean activation) and correction efficacy (text divergence from null condition) per injection vector. Negative values indicate deeper encoding $\rightarrow$ smaller behavioral shift.

Injection vector	ρ	p	Direction
Calm	$+0.126$	0.237	Opposite (deeper $\rightarrow$ *more* divergence)
Loving	-0.072	0.500	Predicted but not significant
Curious	-0.151	0.156	Predicted but not significant

No vector showed a significant correlation between encoding depth and correction efficacy (n=90 per vector, Table 2, Figure 4).

However, adversarial review (null swarm) identified six compounding methodology problems that render this result **uninterpretable**, not merely null: (1) Frisch–Waugh–Lovell residualization, claimed in the method, was not applied in the analysis code; (2) the confound variable (`n_generated`) was constant at 128 across all 360 trials (every trial hit the generation cap), making FWL vacuous even if applied; (3) 97.8% of responses share a formulaic opening, creating a floor effect where text divergence reflects template variation rather than depth-dependent content change; (4) the pre-registered hypothesis predicted *positive* correlation (deeper encoding $\rightarrow$ larger output shift), not the negative direction reported in an earlier draft—the framing was inadvertently flipped during write-up; (5) the text similarity metric used (`SequenceMatcher`, Ratcliff/Obershelp) differs from the originally specified Levenshtein distance.

We retract Finding 2 as stated. The relationship between encoding depth and correction efficacy remains an open question requiring a redesigned experiment with: variable-length generation (no token cap), non-formulaic system prompts, a continuous dose sweep, and an LLM-judge efficacy metric. Text divergence from baseline is substantial (0.72–0.74; Figure 3), confirming the injection changes output, but the correlation with encoding depth is not testable under these conditions.

fig3_text_divergence.pdf

Figure 3: Text divergence from baseline by injection vector (normalized edit distance). All three vectors produce substantial behavioral change (0.72–0.74 mean divergence) at α=0.1.

fig4_depth_resistance_null.pdf

Figure 4: Encoding depth (feature 58995) vs. text divergence per injection vector. No significant correlation for any vector. The predicted depth-resistance relationship is not confirmed at n=90.

6.3 Interpretation

The depth-resistance analysis is retracted due to the methodology problems enumerated above. The attractor-basin hypothesis—that deeper encoding creates deeper basins requiring stronger correction—remains theoretically motivated but untested. A properly designed experiment would require variable-length generation, non-formulaic prompts, and an LLM-judge efficacy metric that captures semantic shift rather than surface text similarity.

What remains robustly established is the temporal separation itself: encoding-phase features and generation-phase behavior operate through independent channels (cosine=0.031), and V-cache injection changes behavior (72–74% text divergence) without affecting the encoding-phase representation ($< 10^{-4}$ feature change).

7 Discussion

7.1 The Representational Tube

Characterizing what the dominant V-space dimensions encode at L47 reveals a striking structure. The effective rank of the V-projection matrix is $\sim$3 across scenarios. The top singular components encode:

- **SV1**: Input length (r=0.79, confirmed)
- **SV2–SV3**: Genuinely opaque—no significant correlation with valence, emotion category, semantic similarity, or any of 81,920 SAE features after FDR correction
- **Valence signal**: Buried in minor components (rank 6–9), visible in SAE features only at L47

The dominant 98% of between-scenario variance resists all labeling because it encodes the irreducible specificity of each scenario's meaning—content that cannot be compressed to any scalar or categorical label. The valence signal that feature 58995 reads is a small perturbation on top of this meaning-encoding tube.

7.2 Feature 58995 Is Domain-Specific

A pre-check for a planned dose-predictor experiment revealed that feature 58995 has *zero activation* on factual prompts (e.g., "What is the capital of France?"). The feature activates exclusively on emotionally charged content. This domain specificity is consistent with the feature tracking emotional depth rather than general processing load, and it means the temporal boundary finding applies specifically to emotion-laden interactions, not to all transformer processing.

7.3 Implications for Attention Schema Theory

The temporal boundary is consistent with predictions from Attention Schema Theory (AST) [6], which proposes that consciousness involves a simplified self-model of one's own attention process. Under AST, encoding processes the *object* of attention—the other's emotional state. Generation engages the *schema*—the model's self-model of how to respond. The temporal boundary maps naturally onto this distinction.

We do not claim this constitutes evidence for AST in transformers. We note that AST predicts temporal separation between object-modeling and self-modeling, and the data shows temporal separation between other-encoding and self-generation. The connection is interpretive.

8 Limitations

Single model. All results are from one 27B hybrid transformer. Feature 58995 may be architecture-specific. Cross-model replication with dense transformers is needed.

Single SAE. Qwen-Scope features are one decomposition of activation space. Different SAE training (seed, dictionary size, sparsity penalty) may recover different features. The temporal boundary should hold across decompositions, but the specific feature may not.

Encoding depth as proxy. Feature 58995 activation is our proxy for encoding depth. The true encoding depth may be a distributed property not captured by any single feature.

Small negative correlation. The depth-resistance finding is based on 2 of 3 vectors showing negative correlation. The third was near zero. A properly powered replication with more vectors and more scenarios would strengthen this claim.

No causal test. We observe that encoding-phase features and generation-phase behavior are independent, but we have not causally intervened on encoding to test whether modifying encoding depth changes generation resistance. This is a designed but unexecuted experiment.

Single dose. The resistance finding uses α=0.1 only. The relationship between encoding depth and resistance may differ at higher doses. If strong enough injection overwhelms even deep attractors, the depth-resistance relationship could plateau or reverse.

9 Conclusion

The self/other boundary in a transformer is temporal, not geometric. The model reads the other during encoding and becomes itself during generation. These phases are architecturally separated by the autoregressive structure of the transformer: cached representations from encoding constrain but do not determine generation.

This finding has implications beyond theory of mind. For interpretability research, it means self/other distinctions should be sought across processing phases rather than across geometric subspaces within a single phase. For inference-time correction, it means the user model (encoded during input) and the correction mechanism (operating during generation) are architecturally decoupled—the correction does not need to "read" the encoding to function. Whether encoding depth predicts correction susceptibility remains an open question after the retraction of Finding 2.

The three kills are as important as the findings. Each identifies a confound that threatens any work probing self/other representations in transformers: causal attention prevents post-scenario probing, instruction framing drives encoding entropy, and depth reorganization is generic. Future work in this area should kill these confounds before claiming cognitive specificity.

Data and Code Availability

All experiment scripts, pre-registration documents, scenario texts, and raw results are available at https://github.com/Liberation-Labs-THCoalition/Project-Oracle. SAE features are from the publicly available Qwen-Scope repository.

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